Self-Training-Based Semantic-Balanced Network for Weakly Supervised Object Detection in Remote-Sensing Images

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***Abstract*— A weakly supervised object detection (WSOD) task is to train a detector with only image-level labels provided. Except for the training difficulty introduced by weaker annotations, the inherent complexity of the remote-sensing images (RSIs) also adds to the challenge. To boost the detector's localization accuracy, we aim to exploit more semantic information contained in images and help improve the general robustness of the model. Noticing previous methods tend to focus on the most discriminative part of an object, we design a self-training-based network that leverages local semantic features. To this end, we develop a semantic-balanced localization module (SBLM) that distinguishes foreground from background and accurate proposals from incomplete ones, by leveraging a balance of region of interest (ROI) and its context information. Moreover, we find that the self-training strategy highly relies on the quality of pseudo-ground-truth boxes. Motivated by this possible lack of robustness, we design a comprehensive clustering module (CCM) and saliency-based proposal filtering (SPF) module that select pseudo-ground truth more comprehensively under supervision. To be more specific, CCM aims to reduce the arbitrariness during assigning pseudo-labels by considering multiple categorical vectors simultaneously. Salient object detection (SOD) is applied in the SPF module to help evaluate the quality of the chosen pseudo-ground-truth boxes. The detection performance is significantly boosted with the proposed method. Extensive experiments conducted on the NWPU VHR-10.v2 dataset and the DIOR dataset validate that the proposed model outperforms the previous state-of-the-art methods favorably with an mAP of 64.9% and 28.1%, respectively.**

***Index Terms*— Localization scores, pseudo-ground truth, remote-sensing images (RSIs), saliency map, weakly supervised object detection (WSOD).**

I. Introduction

OBJECT detection is a significant task in remote-sensing images (RSIs) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. With the rapid development of sensor technology, more data is available for training the detector. Driven by this, the object detection task has greatly advanced in recent years. However, as the amount of data increases, there is a growing demand for manually labeled annotations. The labeling process is laborious because subtle annotations for each instance are required in the common fully supervised object detection tasks. To alleviate the problem, a weakly supervised paradigm [13], [14], [15], [16], [17], [18], where only image-level categorical labels are needed, is expansively leveraged to perform the detection task. The paradigm is called weakly supervised object detection (WSOD).

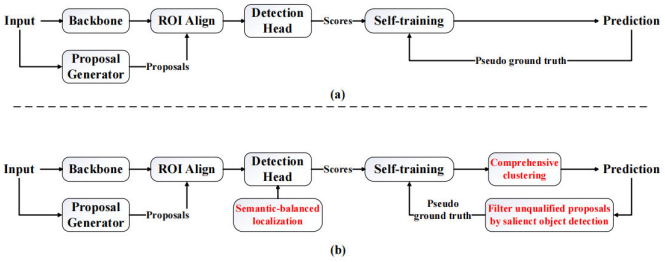


Fig. 1. Flow of MIL with self-training. (a) Original working flow. (b) Flow with our proposed method.

The widest leveraged structure for WSOD methods is multiple instance learning (MIL) [19] shown in Fig. 1(a). In this flow, the image first goes through the backbone. After this, features of regions in proposals are extracted and then forwarded to the detection head as separate inputs. In this way, the detection task is decomposed into the classification problem of different proposals. To better localize the objects, a self-training strategy is always leveraged based on an MIL structure. Under this paradigm, proposals with a great possibility of containing integrated objects are iteratively assigned as pseudo-ground truths that help with the training. This scheme is frequently used in WSOD in RSIs, in which the cluttered background adds to the difficulty of localizing objects accurately.

Even though previous methods have improved performance in WSOD in RSIs, two challenges are yet to be solved. First, with the absence of instance-level labels, it is an inherent problem for WSOD that the detector focuses on the most discriminative part instead of the whole of an object. The

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Fig. 2. Results of different scoring methods. The first and the second images are the results of previous methods, and the third is our method's result. (a) and (b) Results of original methods. (c) Preferable result.

second challenge is the lack of robustness of the pseudo-ground-truth selection process.

Regarding the first challenge, researchers have taken different ways to score the localization quality proposals, but the methods are not complete enough. The original scoring method considers only the inner region of a proposal, resulting in a reverse in scores for integrated ones and proposals covering discriminative parts, as shown in Fig. 2(a). Some other approaches take the information gap between the inner region and outer one of a proposal as the evaluation standard, which may lead to ambiguity within some partly covered proposals and background proposals, as shown in Fig. 2(b).

The second challenge exists in the self-training process. In the previous flow shown in Fig. 1(a), once the pseudo-ground-truth boxes are selected, they serve as the labeling standard for other proposals. As such, an incomprehensive analysis of the scores may lead to general inaccuracy of pseudo-labels. This insufficiency of label quality can hardly be discovered later due to the absence of assessment. The inaccuracy thus continues to exist in every iteration and leads to the futility of the self-training flow.

To address the first problem, a novel semantic-balanced localization module (SBLM) is designed to facilitate localization by further exploiting the context information of proposals. The diverse substandard conditions shown in Fig. 2 motivate us to leverage both inherent and context information of a proposal and form an ideal scoring method. Fig. 2(c) is preferable, in which proposals with dropping localization accuracy have consistent dropping scores. To achieve this, we first stress the dominant role of the feature inside a proposal, since it is the basic standard for distinguishing objects from the background. Then, for a considered foreground proposal, information discrepancy between its inner and outer regions is leveraged to measure the integrity of the covered object. Conclusively, collaborating the inherent semantic knowledge of an object and its saliency to surroundings helps formulate our approach. A balance of the local semantic information is gradually achieved during training.

To tackle the second challenge, we propose an approach with two parts: a comprehensive clustering module (CCM) and saliency-based proposal filtering (SPF). They concentrate on the selection strategy and the postevaluation of pseudo-ground truth, respectively. The new training flow is designed as shown in Fig. 1(b). The novel selection strategy in CCM is motivated by the ambiguity of the previously adopted one, which analyzes proposals by considering their single-categorical score vectors in an arbitrary order. Thinking this may confuse proposal labels, we managed to select candidate pseudo-labels more thoroughly. The SPF is designed to make up for the absence of assessment for assigned labels. Based on our observation of the characteristics of RSIs, saliency object detection [20], [21], [22], [23], [24], [25], [26] is applied to chosen pseudo-ground-truth proposal regions. With the saliency map generated, we can judge how a proposal is occupied by the target object. Some inaccurate proposals can be distinguished in this way. A saliency loss is further introduced to penalize the proposals with poor localization.

A collaboration of these proposed methods formulates our self-training-based semantic-balanced WSOD network, where context information is employed for better localization, and intrinsic characteristics of RSIs are further exploited to enhance the overall robustness. Extensive experiments conducted on the NWPU VHR-10.v2 dataset [46] and the DIOR dataset demonstrate the effectiveness of our proposed methods with an mAP of 64.9% and 28.1%, respectively. Our contributions are as follows.

1) We design a novel self-training-based semantic-balanced WSOD network. The training process becomes more robust and the general detection performance is boosted.
2) SBLM is developed by leveraging the context information of proposals. It facilitates the foreground proposals to stand out from the background ones. Accurate ones can be distinguished from less accurate ones. Proposals that cover integrated objects are more likely to be kept, and the general localization accuracy can be improved in this way.
3) CCM and SPF are introduced to increase the quality of pseudo-ground-truth boxes. CCM helps reduce the ambiguity of the label-assigning process. SPF manages to assess the accuracy of selection by leveraging saliency maps. The effect of the self-training algorithm is more promising with the modules.

II. Related Works

A. Weakly Supervised Object Detection

WSOD is the task where only weak annotations, which are normally image-level labels, are available. Regarding this task, models are usually designed with an MIL structure, which is composed of a proposal generator, a backbone, and a detection head. The proposals are generated in unsupervised ways like selective search (SS) [27], sliding windows (SW), or edge boxes (EB) [28], and the pioneering work of the MIL-based model is WSDDN [29]. It proposes an end-to-end architecture performing region selection and classification simultaneously with two branches in the detection head. However, it exposes a problem that the bounding boxes tend to cover the most discriminative part instead of the whole object.

The problem is ameliorated in later proposed OICR [30] and PCL [31]. They share the same structure with newly designed instance refinement layers that can refine bounding boxes iteratively. Comparatively, PCL is superior to OICR because it adopts a novel proposal label assignment strategy, which improves the quality of pseudo-ground-truth boxes during self-training-based refinements. Self-training strategy is leveraged in the refinement process. This strategy learns a

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representation of the unlabeled data and provides supervision for itself automatically. Many previous works leverage this method to improve general detection performance. SimCLR [32] and MoCo [33] initialize object detection backbones with unsupervised representations learned on off-the-shelf datasets instead of supervised ones. Detreg [34] introduces a new self-supervised method that pretrains the entire object detection network, including the object localization and embedding components. Wei et al. [35] advocated a design principle that encourages alignment between the self-supervised pretext task and the downstream task. SSM [36] proposes a principled self-supervised sample mining process accounting for the real challenges in object detection. To further improve the localization accuracy in WSOD tasks, MELM [37] proposes a min-entropy latent model optimized with a recurrent learning algorithm. ICMWSD [38] employs instance-aware self-training that adds bounding box regression. CASD [39] enforces consistent object detection across different transformations of the same images by computing comprehensive attention and conducting self-distillation on the WSOD networks. Except for self-training algorithms, other strategies like leveraging contextual information are also adopted in the methods above. However, these natural-scene-based methods cannot be directly applied to RSIs because they are usually larger in scale and have a more complicated background. With the great disparity in the domain, some methods are proposed specially targeted for WSOD in RSIs.

B. WSOD in RSIs

Han et al. [40] first tried to address the problem of WSOD in RSIs by proposing a novel and effective geospatial object detection framework that can better mine high-level features progressively. After WSDDN came out, many WSOD methods in RSIs followed its basic structure to boost the overall performance. Yao et al. [41] proposed a dynamic curriculum learning strategy to gradually train the detector with samples of increasing difficulty, which is measured by local entropy. Given the inherent complexity of RSIs, localization ambiguity becomes a more salient challenge. Motivated by this, TCANet [42] introduced a global context-aware enhancement (GCAE) module and a dual-local context residual module to better employ the context information of the images. With these modules applied, proposals can better focus on the whole region of an object instead of its most discriminative part.

Some works managed to employ the peculiarity of RSIs compared with natural-scene images. Based on the fact that more than one instance may exist in a certain class in an RSI, Wang et al. [43] designed a unique multiple instance graph (MIG) learning framework that can find more credible high-quality objects. Besides, objects with apparent similarity are considered to belong to the same class. Another observation is that detectors may face inconsistency across different spatial transformations of the same RSI. To tackle this challenge, an adversarial dropout–activation block and a self-supervised transformation equivariance mechanism are introduced in SAENet [44]. The same challenge is also targeted in RINet [45], which designed a rotation-invariant multiple instance mining strategy coupling instances from different rotation-perceptive branches in a complementary manner. This further improved the performance of the detector. However, there is a lack of robustness during self-training with previous methods adopted. In terms of this, our proposed method optimizes the self-training flow and further improves general localization accuracy.

III. Proposed Methods

The general architecture of our network is shown in Fig. 3. Our network manages to alleviate the localization inaccuracy by simultaneously employing a novel localization module and improving the pseudo-label assigning flow.

A. Overall Structure

We select a self-training-based MIL model as our basic structure, which contains a backbone, detection head, and refinement layers. Proposals are generated by the SS method and are sent to the network along with the input images.

Given an input image I, its feature map can be generated by it going through the backbone network. Then the map is forwarded to the region-of-interest (ROI)-align layer along with the proposals to extract features of corresponding proposal regions. These region features serve as input for the detection head, which is composed of two parallel streams, where we get the two score matrixes M^l and M^c both of size $R \times C$. R is the number of proposals and C is the number of categories. Since the matrices are normalized along the proposal dimension and the category dimension, the obtained S^l and S^c can each represent the score for localization and the score for classification. The final score matrix Φ, each element of which means the probability of a proposal belonging to a category, is calculated by their Hadamard product, as follows:

$$\Phi_{rc} = S^c_{rc} \odot S^l_{rc}. \tag{1}$$

And the score of a category existing in an image Ψ is calculated by the sum of Φ along the proposal dimension

$$\Psi_c = \sum_{r=1}^{R} \Phi_{rc}. \tag{2}$$

With image-level labels $\mathcal{Y} = [\mathcal{Y}_1, \ldots, \mathcal{Y}_C]$ with the value of $\{-1, 1\}$ provided, a vector showing the image categories can be obtained. The classification loss is formulated as below

$$\text{Loss}_{\text{cls}} = \frac{1}{C} \sum_{c=1}^{C} \max(0, 1 - \mathcal{Y}_c \cdot \Psi_c). \tag{3}$$

The instance refinement layers consisting of several fully connected layers are at the end of the structure. They are employed to better localize the objects by propagating higher scores to proposals covering a larger part of the objects iteratively. More specifically, with scores of the proposals calculated through the former network, some top-scoring proposals are assigned as pseudo-ground-truth boxes. Then, labels can be assigned to other proposals according to their overlapping ratio with the pseudo-ground-truth boxes. Refinement

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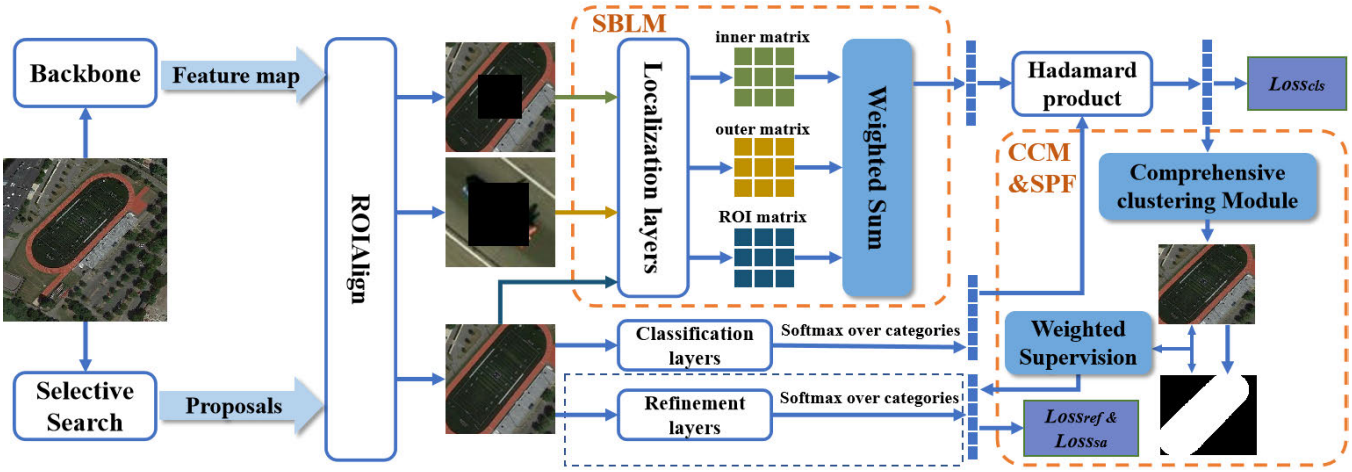


Fig. 3. Architecture of the proposed method. In the main flow, regional features are extracted after the ROI align module. They are forwarded to our proposed SBLM module marked in yellow along with their inner and outer regions to generate scores for regions. During the self-training process, the scores serve as the original standard for pseudo-ground-truth selection, where CCM is applied for better analysis. After selecting the pseudo-ground-truth boxes, SPF is leveraged to further improve their quality by adding $Loss_{sa}$. The model is optimized iteratively in this way.

detectors can be trained under the supervision of these pseudo-labels. The refinement loss is defined as

$$\text{Loss}_{\text{ref}} = -\frac{1}{R}\sum_{r=1}^{R}\sum_{c=1}^{C+1}\hat{y}\log y. \tag{4}$$

$\hat{y}$ is the pseudo-ground truth and y is the predicted score.

On this basis, we propose the SBLM, where the ROI and context features collaborate to distinguish the target object. In terms of pseudo-label assigning, a more comprehensive clustering technique is leveraged, and regional saliency maps are generated to help improve labeling quality. The two ideas each form the CCM and SPF. Moreover, a saliency loss is proposed in SPF to facilitate training. All these methods are designed based on our observation of RSI characteristics. They help the role of each other and form a close loop in optimization. These modules will be detailed in Sections III-B–III-D.

B. Semantic-Balanced Localization Module

Better localizing the objects has always been a challenge for WSOD methods. Often, a bounding box can cover only the most discriminative part of an object. To this end, TCANet [42] introduced an affinity-context block and an ambient-context block to extract features of the inner region and outer region of a proposal respectively. Based on the assumption that the affinity context and ambient context should vary largely in semantic information for a well-localized proposal in RSIs, the original localization score is substituted with the subtraction of the affinity context score, and the ambient context score generated by the same fully-connected layers. This is reasonable because there is supposed to be a shift in semantic features while crossing the boundary of an object.

With this detection score applied, however, the residual of the affinity context and ambient context scores makes sense while either of the two scores becomes nonsensical individually. This presents an issue that we overemphasize the information gap between the inner and outer regions of a proposal, yet the inherent value of an ROI is ignored. This violates the fundamental law of scoring because the information discontinuity at proposal boundaries is not equal to accurate localization in all cases.

To be more specific, we can discuss the latent rules for the ideal scoring function. It should assign different-level scores to different cases. A perfectly localized proposal can earn the highest score with no doubt. The score should be relatively lower for proposals that are overly large or cover only part of the target. As for the proposals fully occupied with background, the score is expected to be close to zero. The residual localization score, however, may treat some proposals except for the perfectly localized ones equally. Because in the last two cases, information on the proposal boundaries tends to be continuous.

We consider it essential for a qualified localization module to perform the aforementioned laddering scores. To achieve this, distinguishing foreground objects from background ones should be dominant. Given the conventional relatively low scores of accurate proposals compared with proposals covering part of an object, a reversal of the scores in the two conditions is also needed. We observe that objects tend to be nonoverlapping in RSIs, so the semantic information gap between features on two sides of the boundary of accurate proposals is leveraged as an indication of an integrated region.

Practically, the ROI region, the inner region, along with the outer region of a proposal are all extracted by ROI-align at first. They are then forwarded to the same fully connected layers to generate separate score matrixes for localization, namely ROI score matrix M^{roi}, inner score matrix M^{in}, and outer score matrix M^{out}, respectively. Prompted by previous methods, the information gap δ^{loc} in range 0–1 is quantitatively calculated by the residual of M^{in} and M^{out}, and with ROI information considered, the new localization score matrix M^{l} can be formulated as

$$M^{l} = M^{\text{roi}}_{rc} + \lambda_{\text{loc}}\left(M^{\text{in}}_{rc} - M^{\text{out}}_{rc}\right) \tag{5}$$

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where λ_{loc} is a hyperparameter. The final localization score S^l is as below

$$S_{rc}^{l} == \frac{e^{M_{rc}^{l}}}{\sum_{r=1}^{R} e^{M_{rc}^{l}}}. \tag{6}$$

As we expected before, the ROI score is considered. When the object is correctly localized, subtraction is a higher value, raising the whole score. In other cases, it contributes little, thus the inherent distinction between foreground and background proposals can play the dominant role. A quantitative demonstration of the effectiveness of this module is given in Section IV-E.

C. Comprehensive Clustering Module

Despite the novelty of the clustering method, we do not perceive it as flawless. With the image-level label provided, it is common practice to consider existing categorical score vectors separately in an arbitrary sequence, meaning the latter categories will ignore vectors of the previously determined proposals. This may bring about misclassification, especially for proposals with ambiguous scores. In other words, a proposal with a relatively high probability of belonging to different categories is supposed to be judged more comprehensively.

As such, we propose to perform the clustering according to a scoring matrix of all the possible categories of proposals. In other words, the input data for clustering is of size $R \times C_e$, where R is the number of proposals and C_e is the number of existing categories. Since mean-teacher clustering is employed, a hyperparameter is needed for the number of clusters. In our method, we assign it as the number of categories: the foreground types involved in the image along with the background type. In the ideal case, in which each cluster has the highest average score of a distinct category, we can assume that a certain cluster is a group of candidate proposals for a certain category. The group of background proposals can be eliminated. However, it happens that one cluster may be ambiguous with the highest average score in multiple or no categories at the beginning. We adopt the former clustering method of considering categories separately under this condition. However, the problem can be alleviated as training progresses since there is less randomness.

A sample of applying our proposed clustering method is shown in Fig. 4, where three clusters are included, indicating that there are two foreground categories in the image. The two axes represent scores of proposals belonging to each of the categories respectively. Intuitively, considered background proposals marked in red can be eliminated. Foreground proposals gathering at the boundary of the blue and the green clusters are classified according to scores of multiple dimensions. However, with the former strategy adopted, these proposals will likely be placed into the category analyzed first.

After the clustering process, we further restrict the number of candidate ground-truth boxes for each category to less than 5, in an attempt to ensure the quality of the candidate boxes. This is accomplished by performing iterative clustering in candidates of the same category until the requirement is met.

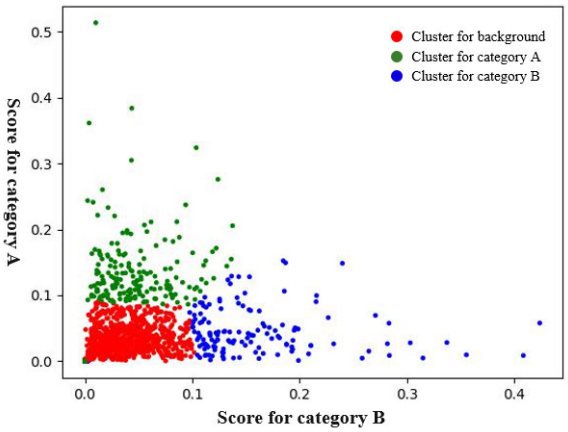


Fig. 4. Clustering result of our proposed method. The red points represent the considered background cluster. The other two colors represent clusters of existing foreground categories.

This also helps in improving the general quality of pseudo-ground-truth boxes.

D. Saliency-Based Proposal Filtering

In our framework, after the top-scoring proposals are selected as pseudo-ground-truth boxes, they serve as the standards for labeling the other proposals. Practically, proposals with IoU larger than a threshold with any of the pseudo-ground-truth boxes are assigned the labels of their max-overlapping pseudo boxes. While those with IOU lower than another preset threshold are considered to be background boxes. The labels then act as supervision for iterative refinements. So, it can be inferred that the model's learning outcomes highly rely on the quality of the pseudo-ground-truth boxes. Noticing their significance, we seek approaches to penalize false-positive pseudo-ground truth by further exploiting features of RSIs.

It is observed that in RSIs, objects are usually nonoverlapping, which means the bounding boxes tend to be separate. Based on this assumption, a preferred proposal is likely to enclose only the target object and avoid any part of the others. As such, the foreground region inside a proposal is supposed to belong to a single instance. Moreover, if the instance is rather accurately localized, it should be placed at the center of the box. To tell whether the desirable conditions are met, salient object detection (SOD) can be applied to each proposal to distinguish the foreground part and further analyze its location.

Given that the saliency map in a proposal is associated with localization performance to some degree, a proposal selection criterion built upon SOD on regions in pseudo-ground-truth boxes is designed. The core of our method is to measure the distance between the saliency center c_s and the geometric center c_g of a proposal. The closeness between the centers is considered a requisite for accurate localization. The specific working process is as follows.

After label assigning, we extract regions in the pseudo-ground-truth proposals as separate images, where Gaussian denoising is first applied to restrain the noise of the background. Then the SOD model pretrained on another off-the-shelf remote-sensing dataset is leveraged to get the corresponding saliency maps, of which the center of mass

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Fig. 5. First row includes sample proposal regions, and their corresponding SOD results are shown in the second row. The red and cyan centers in saliency maps represent the geometric centers and saliency centers, respectively. The line connecting the two points in a figure is the absolute Euclidean distance between them.

is calculated, namely the saliency center. The center of the proposal is called the geometric center. With the coordinates of the two points, the normalized Euclidean distance D_r between them can be calculated as

$$D_r = \frac{2\|c_s - c_g\|}{\sqrt{a_r^2 + b_r^2}} \tag{7}$$

where a_r and b_r are the length and width of the proposal, respectively. Fig. 5 perceptually illustrates this measurement by showing regions in proposals and their corresponding saliency maps, where the geometric centers and saliency centers are marked in red and cyan, respectively. The line connecting the two points in a figure is the absolute Euclidean distance between them. So, the length of the lines perceptually shows the closeness of the two centers.

In the accurate proposals, the lines are nearly indiscernible. While an apparent distance between the two centers is a sign of being inaccurate. Noticing a larger distance suggests a greater deviation, we propose a loss that is directly proportional to D_r of a pseudo-ground-truth box. The loss function is formulated as below

$$\text{Loss}_{\text{saliency}} = \frac{1}{N_{\text{pgt}}} \sum_{m=1}^{N_{\text{pgt}}} l(D_m) e^{D_m}. \tag{8}$$

In this function, N_{pgt} is the number of pseudo-ground-truth proposals. The loss has a range of 0–e. An exponential function is leveraged since it is convex. So, there is an increased gradient with the increase of D_r. This suggests that an overly large D_r will be more severely punished. A step function $l(x)$ as (9) is employed for the same purpose, which is to give some tolerance to D_r of accurate proposals. The threshold λ_r belongs to the constant set $\mathbf{C}$. λ_r is set to 0.5

$$l(x) = u(x) = \begin{cases} 1, & x \le \lambda_r \\ 0, & x > \lambda_r \end{cases}, \quad \lambda_r \in \mathbf{C}. \tag{9}$$

The final loss for the whole model is formulated as below. The saliency loss is added to the total loss, indicating that the quality of pseudo-ground-truth boxes will be gradually improved according to the saliency maps. Scores generated by the detection head can get the feedback and thus be optimized. Conclusively, the SPF module formulates a saliency loss and acts as the final part of the training flow. It improves the performance of the model by simultaneously influencing the scoring and refinement process

$$\text{Loss}_{\text{total}} = \text{Loss}_{\text{cls}} + \text{Loss}_{\text{ref}} + \text{Loss}_{\text{saliency}}. \tag{10}$$

IV. Experiments

A. Datasets and Evaluation Metrics

We adopt the commonly used datasets of previous WSOD methods in RSIs for a fair comparison. Either the NWPU VHR-10.v2 dataset [46] or the DIOR dataset [47] is leveraged for training and testing. The NWPU VHR-10.v2 is a ten-category dataset that contains 1172 images of 400 × 400 pixels. It is split into a training set of 879 images and a test set of 293 images. The DIOR dataset is comparatively large with 23 463 800 × 800 sized images that contain 20 categories. The dataset is divided into a train set, a test set, and a validation set. The training and validation sets are merged into a trainval set of 11 738 images for our training. It is worth noting that only image-level category labels of the datasets are leveraged in our experiments.

Two metrics for WSOD are employed for evaluation: mean average precision (mAP) and correct localization (CorLoc). Average precision (AP) is the weighted average of precision under different score thresholds. mAP is calculated with the mean of AP of all the categories. This metric is used to measure the overall detection performance on test sets. CorLoc is a metric for localization accuracy. It is the percentage of images with all targets correctly localized by the top-1 prediction of a certain category. Following the setting in previous methods, mAP is applied on the test set while CorLoc is applied on the trainval set.

B. Implementation Details

The architecture of our network is based on an MIL model with a VGG16 backbone, which is initialized by ImageNet pre-trained parameters. Since the self-training strategy is adopted, instance refinement layers are added to the model like OICR [30]. The number of refinements is set to 2. Following the setting in TCANet [42], The number of overall iterations for the NWPU VHR-10.v2 dataset and the DIOR dataset is set to 30k and 200k, respectively. The basic learning rate for both is 0.001, which will decay by ten times after 10k and 100k iterations. In SOD, a detector [20] already trained on the ORSSD dataset is applied and its weights are set to be frozen.

C. Comparison With SOTA

We compare the proposed method with previous state-of-the-art methods in WSOD in this section. Since the MIL-based model is similar to Fast-RCNN [48] in general structure, some fully supervised approaches are also considered. Tables I and II present mAP and CorLoc performance results when NWPU VHR-10.v2 is adopted as the training set. In terms of mAP, our method surpasses the previous best WSOD method by 2.2%. The result exceeds 60.7% of SAENet [44] by more than 4% and 58.8% of TCANet [42] by more than 6%. Especially, the mAP performance

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TABLE I

MAP COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE NWPU VHR-10.V2 TEST SET

Methods	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	mAP
Faster RCNN [48]	90.9	86.3	90.5	98.2	89.7	69.6	100.0	80.1	61.5	78.1	84.5
WSDDN [29]	30.1	41.7	35.0	88.9	12.9	23.9	99.4	13.9	1.9	3.6	35.1
OICR [30]	13.7	67.4	57.2	55.2	13.6	39.7	92.8	0.2	1.8	3.7	34.5
PCL [31]	26.0	63.8	2.5	89.8	64.5	76.1	77.9	0.0	1.3	15.7	39.4
MELM [37]	80.9	69.3	10.5	90.2	12.8	20.1	99.2	17.1	14.2	8.7	42.3
TCANet [42]	89.4	78.2	78.4	90.8	35.3	50.4	90.9	42.4	4.1	28.3	58.8
SAENet [44]	82.9	74.5	50.2	96.7	55.7	72.9	100.0	36.5	6.3	31.9	60.7
RINet [45]	90.4	87.1	43.7	93.3	36.8	63.0	100.0	45.7	18.2	49.1	62.7
Ours	88.3	84.5	71.7	95.4	30.9	66.7	100.0	46.9	23.3	41.0	64.9

TABLE II

CORLOC COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE NWPU VHR-10.V2 TRAINVAL SET

Methods	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	CorLoc
WSDDN [29]	22.3	36.8	40.0	92.5	18.0	24.2	99.3	14.8	1.7	2.9	35.2
OICR [30]	29.4	83.3	20.5	81.8	40.9	32.1	86.6	7.4	3.7	14.4	40.0
PCL [31]	11.8	50.0	12.8	98.7	84.5	77.4	90.7	0.0	9.3	15.6	45.1
MELM [37]	86.0	77.4	21.4	98.3	10.7	43.5	95.0	40.0	11.8	14.6	49.9
TCANet [42]	96.9	91.8	95.1	88.7	66.9	62.8	96.0	54.2	19.6	55.6	72.8
SAENet [44]	97.1	91.7	87.8	98.7	40.9	81.1	100.0	70.4	14.8	52.2	73.5
RINet [45]	91.2	90.3	64.3	100.0	50.0	82.6	100.0	50.0	41.2	65.9	73.5
Ours	94.7	90.3	42.9	100.0	53.6	87.0	100.0	70.0	47.1	73.2	75.9

TABLE III

MAP COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE DIOR TEST SET

Methods	PL	AP	BF	BC	BR	CM	DA	ES	ET	GF	GTF	HB	OP	SH	SD	ST	TC	TS	VH	WM	mAP
FastRCNN	44.2	66.8	67.0	60.5	15.6	72.3	52.0	65.9	44.8	72.1	62.9	46.2	38.0	32.1	71.0	35.0	58.3	37.9	19.2	38.1	50.0
WSDDN [29]	9.1	39.7	37.8	20.2	0.3	12.1	0.6	0.7	11.9	4.9	42.4	4.7	1.1	0.7	63.0	4.0	6.1	0.5	4.6	1.1	13.3
OICR [30]	8.7	28.3	44.1	18.2	1.3	20.2	0.1	0.7	29.9	13.8	57.4	10.7	11.1	9.1	59.3	7.1	0.7	0.1	9.1	0.4	16.5
PCL [31]	21.5	35.2	59.8	23.5	3.0	43.7	0.1	0.9	1.5	2.8	56.4	16.8	11.1	9.1	57.6	9.1	2.5	0.1	4.5	4.6	18.2
MELM [37]	28.1	3.2	62.5	28.7	0.1	62.5	0.2	13.1	28.4	15.2	41.1	26.1	0.4	9.1	8.6	15.0	20.6	9.8	0.0	0.5	18.7
TCANct [42]	25.1	30.8	62.9	40.0	4.1	67.8	8.1	23.8	29.9	22.3	53.9	24.8	11.1	9.1	46.4	13.7	31.0	1.5	9.1	1.0	25.8
SAENet [44]	20.6	62.4	62.7	23.5	7.6	64.6	0.2	34.5	30.6	55.4	52.7	17.6	6.9	9.1	51.6	15.4	1.7	14.4	1.4	9.2	27.1
RINet [45]	34.0	69.2	63.2	42.1	0.5	65.7	0.2	0.0	0.0	43.9	54.0	36.2	0.4	7.8	53.2	17.1	38.3	14.6	9.1	0.7	27.5
Ours	31.0	67.0	63.1	37.8	0.3	71.1	0.2	0.0	0.0	62.3	53.2	40.5	0.6	9.1	53.3	16.6	32.2	13.8	9.1	0.5	28.1

TABLE IV

CORLOC COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE DIOR TRAINVAL SET

Methods	PL	AP	BF	BC	BR	CM	DA	ES	ET	GF	GTF	HB	OP	SH	SD	ST	TC	TS	VH	WM	CorLoc
WSDDN [29]	5.7	59.9	94.2	55.9	4.9	23.4	1.0	6.8	44.5	12.8	89.9	5.5	10.0	23.0	98.5	79.6	15.1	3.5	11.6	3.2	32.5
OICR [30]	16.0	51.5	94.8	55.8	3.6	23.9	0.0	4.8	56.7	22.4	91.4	18.2	18.7	31.8	98.3	81.3	7.5	1.2	15.8	2.0	34.8
PCL [31]	61.1	46.9	95.4	63.6	7.3	95.1	0.2	5.7	5.1	50.8	89.4	42.1	19.8	37.9	97.9	80.7	13.8	0.2	10.5	6.9	41.5
MELM [37]	77.0	28.9	92.7	63.0	13.0	90.1	0.2	17.0	37.9	44.6	88.1	49.4	15.7	28.2	98.3	83.0	22.8	10.3	4.6	2.2	43.3
TCANet [42]	91.2	69.4	95.5	67.5	18.9	97.8	0.2	70.5	54.3	51.4	88.3	48.0	2.3	33.6	14.1	83.4	65.6	19.9	16.4	2.9	49.6
SAENet [44]	91.2	69.4	95.5	67.5	18.9	97.8	0.2	70.5	54.3	51.4	88.3	48.0	2.3	33.6	14.1	83.4	65.6	19.9	16.4	2.9	49.6
RINet [45]	78.7	73.5	87.3	63.4	32.1	69.5	35.8	40.9	64.5	67.8	79.7	35.9	32.8	22.4	73.1	65.4	65.2	43.1	4.3	11.6	52.4
Ours	82.3	61.7	88.7	66.0	31.5	79.1	33.9	31.8	65.4	67.0	81.7	40.3	38.4	24.5	69.5	62.6	70.9	50.9	2.9	13.0	53.1

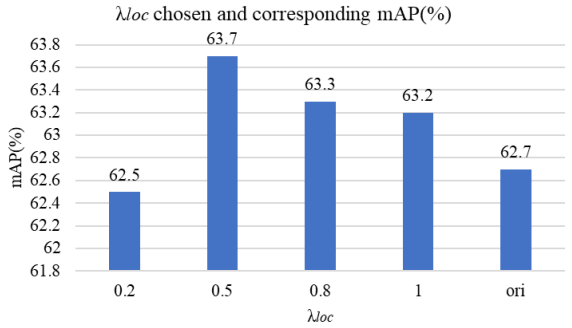


Fig. 6. λ_{loc} chosen for the SBLM module and the corresponding mAP.

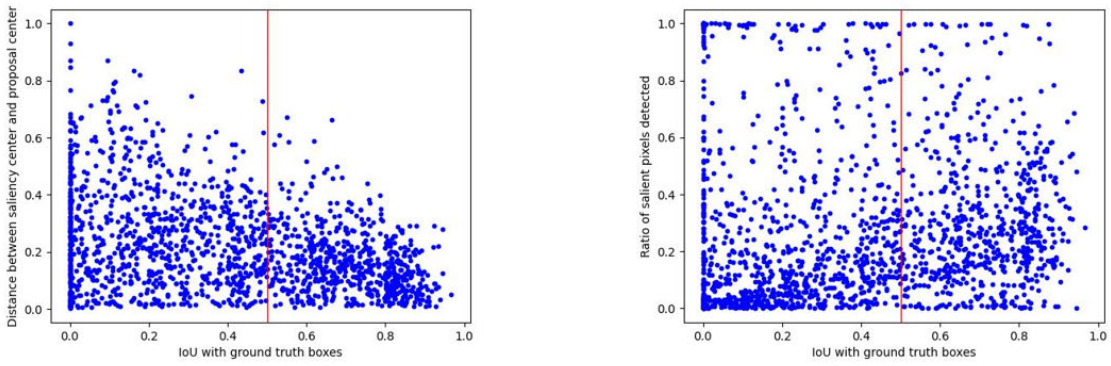
of our proposed method is superior in the Bridge and the Harbor categories, suggesting that we can better deal with nonrigid objects compared with the previous WSOD works. The clear superiority in mAP performance demonstrates the effectiveness of the collaboration of the proposed modules. Meanwhile, our result of CorLoc is 75.9%, gaining a significant improvement compared with 73.5% in RINet [45]. It can be inferred that the performance gap between WSOD and corresponding fully supervised detectors is narrowing with the progress we made.

Results of experiments conducted with the DIOR dataset are shown in Tables III and IV. State-of-the-art performances are also achieved in both mAP and CorLoc metrics. Our proposed approach outperforms the previous best method with a 0.6% gain in mAP and a 0.4% gain in CorLoc. All these quantitative comparisons illustrate the effectiveness of our method.

D. Ablation Studies

1) Components of the Proposed Method: We design SBLM, CCM, and SPF to improve the general performance of WSOD

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(a) (b)

Fig. 7. Statistical results on indicators of the saliency map. The vertical axes of the scatter diagrams represent the indicators we study. The horizontal axes are the IoU with its max-overlapping ground-truth box for a proposal. Since an IoU of 0.5 is chosen as the threshold for correct localization, a red perpendicular line is used to split the points into qualified proposals on its right and unqualified ones on the left. (a) Using the distance between the saliency center and the proposal center as the indicator. (b) Using the ratio of the salient area to the whole area in the proposal as the indicator.

TABLE V
ABLATION STUDY ON COMPONENTS OF OUR METHOD

SBLM	CCM	SPF	mAP (%)
			62.7
✓			63.7
	✓		63.0
		✓	63.5
✓	✓		64.2
	✓	✓	63.9
✓		✓	64.5
✓	✓	✓	64.9

in RSIs. To analyze how they contribute to the overall performance, we conduct experiments on the NWPU VHR-10.v2 test set shown in Table V, where applying each module or not is considered. It is worth noting that a basic OICR structure is adopted when all our proposed methods are not employed. In comparison to this condition, SBLM, CCM, and SPF increase the mAP by 1.0%, 0.3%, and 0.8%, respectively. The collaboration of these proposed approaches can generate even greater improvement. By leveraging SBLM and SPF simultaneously, the general performance can be boosted to 64.5% mAP. Cooperating CCM with either of the other two modules also generates an mAP of 64.2% and 63.9%. With all three modules applied, general mAP can be up to 64.9%, significantly outperforming the basic model.

2) λ_{loc} Chosen for SBLM: In this part, we conduct experiments regarding the value of λ_{loc} in (5). Five numbers are taken into consideration, as shown in Fig. 6. With a different λ_{loc} set, a corresponding mAP performance of the model with only SBLM applied to the basic architecture is leveraged for evaluation. The term "ori" in the table means using the baseline method. We can infer that when λ_{loc} is set to 0.5, the model performs relatively better with an mAP of 63.7%, surpassing the baseline method by 1.0%. Generally speaking, our module is superior to either previous methods of solely leveraging context or ROI information.

3) Indicator of the Saliency Map: In terms of the way of analyzing the saliency maps of different proposals in the SPF module, two indicators are considered at the beginning. The first one, which we finally adopted, is the distance between the saliency center and the proposal center. The indicator works based on the assumption that the localization accuracy is in negative correlation to the distance. The other indicator is the ratio of the salient area to the whole area in the proposal, in which the area is calculated by the number of pixels. A proposal with a large proportion of salient region is preferred, for it is likely to be more fully occupied by a target object. So, if the localization accuracy positively correlates to this ratio, we can associate the model performance with this indicator.

Regarding the two indicators, we statistically analyze their relationships with localization accuracy on the NWPU VHR-10.v2 test set, respectively, as shown in Fig. 7. Each point in the scatter diagrams represents a proposal, whose IoU with its max-overlapping ground-truth box is the corresponding value on the horizontal axis. Since an IoU of 0.5 is chosen as the threshold for correct localization, a red perpendicular line is used to split the points into qualified proposals on its right and unqualified ones on the left. The vertical axes represent the indicators we study. Fig. 7(a) demonstrates our observation that accurate proposals have a relatively small distance between the saliency center and the box center, while Fig. 7(b) is more cluttered. Except for the possible poor performance of SOD, this phenomenon can be explained by the diverse shapes of objects, which lead to the discrepancy in salient area proportion. In comparison, leveraging distance as the indicator can be more robust. The saliency center can be correct even if the foreground and background regions are recognized as the complete opposite. The robustness is further ensured by the design of saliency loss, in which the thresholds help formulate a soft filtering method. With both theoretical and statistical analysis, the distance is finally selected as the indicator for correct localization.

E. Visualization

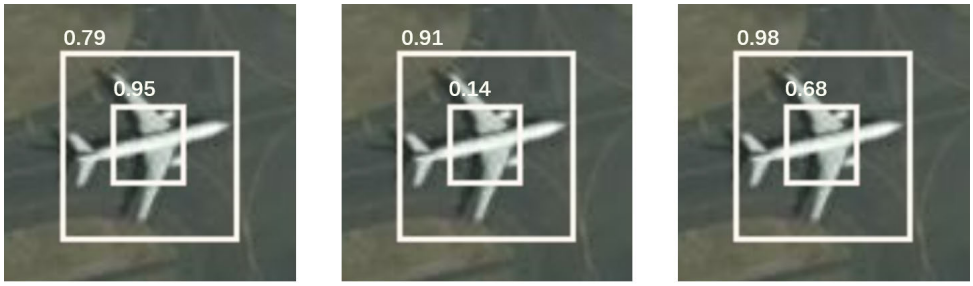
Visualization for SBLM: In the proposed method section, we illustrate the superiority of SBLM theoretically. In this section, sample proposals along with their localization scores under different scoring methods are given in Figs. 8–10 to demonstrate the effectiveness of the module quantitatively. Each figure contains three images showing scoring results of methods that consider ROI only, context information only, and

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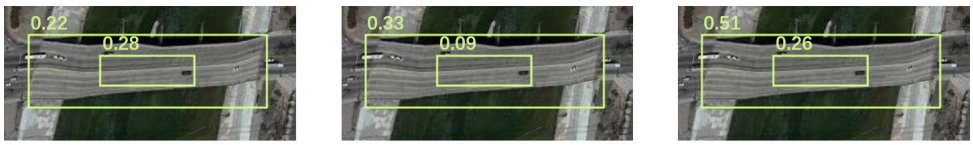
(a) (b) (c)

Fig. 8. Sample 1 for scoring results of three different scoring modules. The larger proposal in an image is accurate, while the smaller proposal is less accurate. Their corresponding scores are written in the top left corners. (a) Method considers ROI only. (b) Method considers context information only. (c) Method considers both ROI and context.



(a) (b) (c)

Fig. 9. Sample 2 for scoring results of three different scoring modules. (a) Method considers ROI only. (b) Method considers context information only. (c) Method considers both ROI and context.



(a) (b) (c)

Fig. 10. Sample 3 for scoring results of three different scoring modules. (a) Method considers ROI only. (b) Method considers context information only. (c) Method considers both ROI and context.

TABLE VI

DETAILED SCORES FOR ALL THE CATEGORIES OF DIFFERENT MODULES IN FIG. 8. FOR EACH MODULE, THE FIRST ROW OF SCORES BELONGS TO THE ACCURATE PROPOSAL

Methods	airplane	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle
(a)	0.06	**0.44**	0.06	0.08	0.12	0.02	0.09	0.05	0.04	0.04
	0.02	**0.52**	0.03	0.16	0.07	0.04	0.06	0.03	0.03	0.04
(b)	0.01	**0.75**	0.00	0.15	0.01	0.02	0.03	0.01	0.01	0.01
	0.08	**0.02**	0.11	0.08	0.10	0.14	0.11	0.11	0.12	0.14
(c)	0.01	**0.82**	0.01	0.07	0.03	0.01	0.03	0.01	0.01	0.01
	0.03	**0.31**	0.05	0.19	0.10	0.07	0.08	0.05	0.04	0.07

TABLE VII

DETAILED SCORES FOR ALL THE CATEGORIES OF DIFFERENT MODULES IN FIG. 9

Methods	airplane	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle
(a)	**0.79**	0.03	0.01	0.01	0.02	0.04	0.02	0.03	0.03	0.01
	0.95	0.00	0.01	0.00	0.01	0.01	0.00	0.01	0.02	0.00
(b)	**0.91**	0.01	0.02	0.01	0.01	0.00	0.01	0.00	0.02	0.01
	0.14	0.06	0.15	0.07	0.09	0.09	0.09	0.10	0.12	0.08
(c)	**0.98**	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.68	0.02	0.06	0.01	0.03	0.04	0.03	0.04	0.08	0.02

both ROI and context, which is our method, from left to right. They correspond to the three conditions we mention in the instruction part. And the detailed scores for all the categories are presented in Tables VI–VIII. Distinctively, our proposed method manages to assign an accurate proposal a higher score, while not mistaking a less accurate one for background. Taking Fig. 8, for example, our scores for the accurate and inaccurate proposals are 0.82 and 0.31, respectively.

The detailed scores for all the categories are shown in the three images in the table below, with the target category

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TABLE VIII
DETAILED SCORES FOR ALL THE CATEGORIES OF DIFFERENT MODULES IN FIG. 10

Methods	airplane	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle
(a)	0.05	0.13	0.05	0.03	0.06	0.05	0.25	0.07	**0.22**	0.10
	0.03	0.10	0.03	0.03	0.08	0.06	0.23	0.10	**0.28**	0.06
(b)	0.07	0.04	0.08	0.05	0.12	0.04	0.15	0.04	**0.33**	0.07
	0.07	0.12	0.07	0.11	0.13	0.11	0.07	0.14	**0.09**	0.08
(c)	0.03	0.04	0.03	0.01	0.05	0.02	0.26	0.02	**0.51**	0.05
	0.02	0.13	0.02	0.04	0.11	0.07	0.16	0.14	**0.26**	0.05

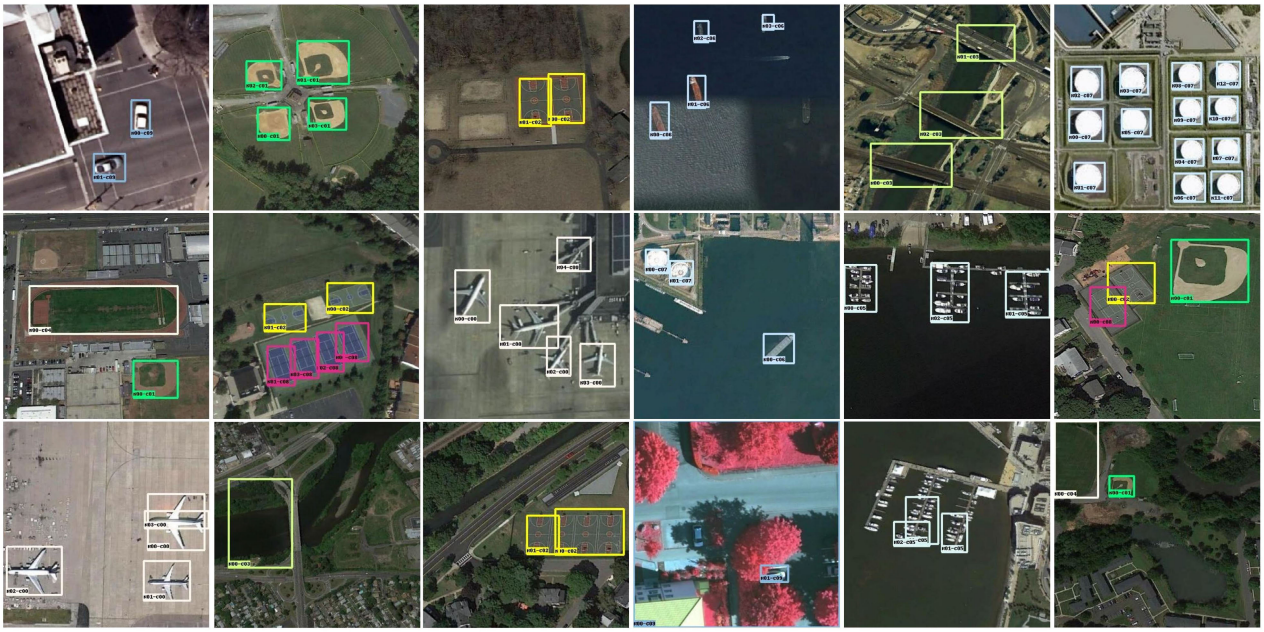


Fig. 11. Detection results on the NWPU VHR-10.v2 test set. The first two rows are successful samples and the third row shows inaccurate results. Category indices are labeled on the top left corner, and their corresponding category names can be found in Table IX in the Appendix.

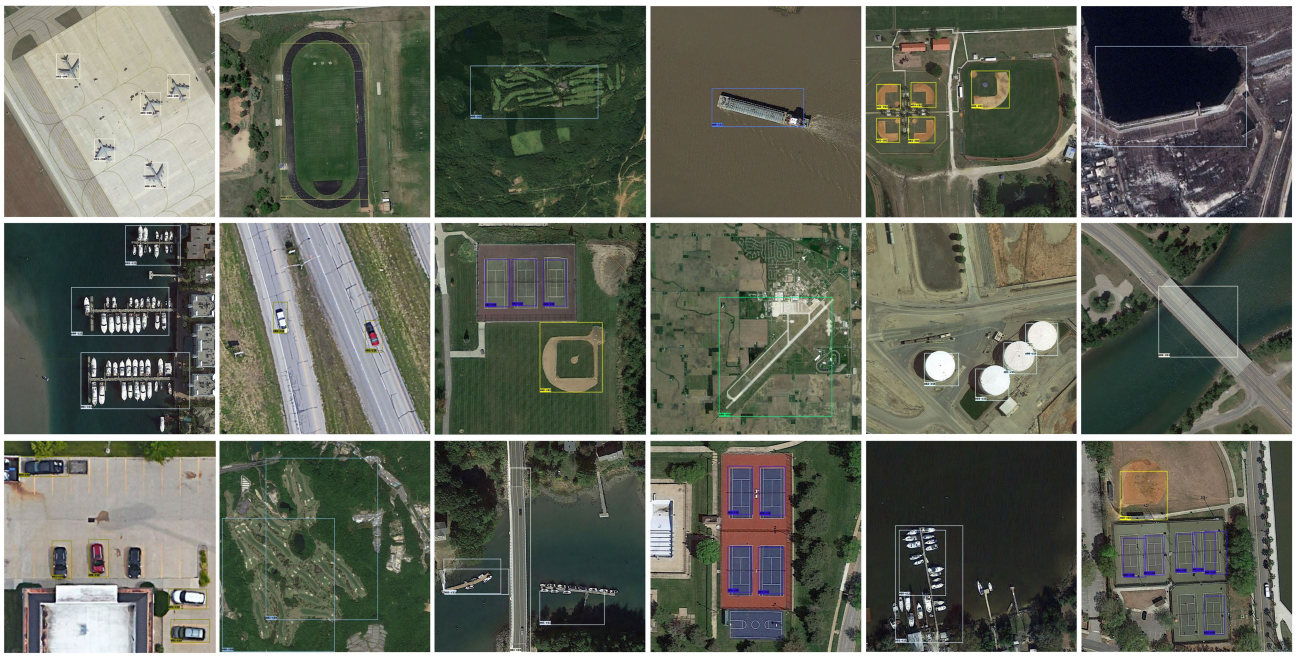


Fig. 12. Detection results on the DIOR test set. The first two rows are successful samples and the third row shows inaccurate results. Category indices are labeled on the top left corner, and their corresponding category names can be found in Table X in the Appendix.

marked in bold. Our baseline method introduces the severe problem that an incomplete object gains an extremely low score, as the scores in Figs. 8(b), 9(b), and 10(b). We observe from Fig. 8(b) that this score of 0.02 is even lower than the score of any of the other categories, which means the score is suppressed too hard by the continuity of feature at proposal

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TABLE IX
INDICES AND CORRESPONDING CATEGORIES FOR VISUALIZATION OF THE NWPU VHR-10.v2 DATASET IN FIG. 11

C00	C01	C02	C03	C04
Airplane	Baseball Diamond	basketball court	Bridge	Ground track field
C05	C06	C07	C08	C09
Harbor	Ship	Storage	Tennis Court	Vehicle

TABLE X
INDICES AND CORRESPONDING CATEGORIES FOR VISUALIZATION OF THE DIOR DATASET IN FIG. 12

C00	C01	C02	C03	C04
Airplane	Airport	Baseball field	Basketball court	Bridge
C05	C06	C07	C08	C09
Chimney	Dam	Expressway service area	Expressway toll station	Golf field
C10	C11	C12	C13	C14
Ground track field	Harbor	overpass	Ship	Stadium
C15	C16	C17	C18	C19
Storage tank	Tennis court	Train station	Vehicle	Wind mill

boundaries. This problem is greatly alleviated with our proposed method applied, as scores of Figs. 8(c), 9(c), and 10(c).

Detection Results Some detection results on the NWPU VHR-10.v2 test set and the DIOR test set are given in Figs. 11 and 12. Generally, our model can distinguish multiple objects existing in the same image from its cluttered background. Objects of different categories are hardly mistaken for each other. But still, the model cannot localize well in some difficult cases, like when the instances are nonrigid. In these cases, the bounding boxes may cover only part of an object or mistake some objects for background. More strategies are to be discovered to alleviate this inherent challenge of WSOD. There will be a smaller performance gap between WSOD and fully supervised methods in the future.

V. CONCLUSION

We design a novel architecture for WSOD on aerial images in this article. To mitigate the problem of unreasonable localization scoring, we developed SBLM that helps the model better distinguish foreground objects and focus on integrated ones as well. In addition, CCM and SPF are proposed to improve the quality of pseudo labels used in self-training. Characteristics of RSIs are further exploited to facilitate our design. Theoretical and statistical analyses both demonstrate the effectiveness of our model. Comprehensive experiments were also conducted and showed significant improvement against the previous state-of-the-art method.

APPENDIX
CORRESPONDENCE OF INDICES AND NAMES FOR IMAGE CATEGORIES IN FIGS. 11 AND 12

See Tables IX and X.

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